MSAI 605 Milestone 2 Report

Face Verification Evaluation Loop (LFW)

By Arun Kulkarni & Pramod Goyal

Section 1: System Overview

The task assigned is face verification on the Labeled Faces in the Wild (LFW) dataset: given a pair of face images, decide whether they depict the same person (label = 1) or different people (label = 0).

The Milestone 1 verifier operates directly on raw pixel intensities; each 62×47 grayscale image is flattened to a 2914-dimensional vector. No learned feature extractor is used. Similarity between a pair is computed as cosine similarity between the two flat vectors.

Cosine similarity is in [−1, 1]; higher values indicate more similar faces. The system predicts the same *person* when `score ≥ threshold` and a *different person* otherwise. (Euclidean distance, if used, would be negated first so the same decision rule applies throughout.) This convention is documented in `configs/eval_config.json` under `score_direction`.

Splits were assigned at the identity level (via `--val-fraction 0.15`, seed 42) to prevent identity leakage across splits (see Appendix, Table 1). The val split was used exclusively for threshold selection; no test data was seen until the threshold was fully locked.

The configuration is as follows, found in `configs/eval_config.json`: threshold grid = [0.0, 1.0] in 200 equal steps, seed = 42, selected threshold rule = `balanced_acc`.

Section 2: Threshold-Selection Procedure

The threshold grid spans [0.0, 1.0] in 200 equal steps. All three candidate rules were evaluated on the validation split only (see Appendix, Table 2).

The AuROC is 0.6109, as shown in Figure 1 (Appendix). The chosen rule was to maximize balanced accuracy, leading to a threshold of 0.9497.

The validation split is imbalanced (35% positive / 65% negative), so balanced accuracy, which weights TPR and TNR equally, is the most appropriate single-number criterion. Among the three rules, it produced the highest balanced accuracy on the val split (0.5899), confirming it as the best operating point for this dataset.

The ROC AUC of 0.6109 is modest, which is expected: raw pixel cosine similarity has limited discriminative power without a learned feature extractor, and any pose or lighting change between two images of the same person can severely reduce their cosine similarity.

Section 3: Selected Threshold and Confusion Matrices

The validation and test run results, including confusion matrices, can be seen in Figure 2 and Figure 3 (Appendix), respectively.

The locked threshold is 0.9497; this was selected on the validation split, and locked before any test evaluation. Given this, the system is heavily biased toward **rejecting** pairs (FRR = 59%). The high threshold of 0.9497 means only pairs with near-identical pixel patterns are accepted as same-person: genuine pairs separated by pose change, lighting, or aging are frequently missed. The test balanced accuracy (0.614) is slightly higher than on val (0.590), suggesting the threshold generalised reasonably well despite the val imbalance. The gap between FAR (18%) and FRR (59%) reflects that the baseline is more conservative than balanced, consistent with a pixel-level similarity measure that degrades under real-world appearance variation.

Section 4: Error Analysis

Two error slices were defined on the baseline test split (1000 pairs, threshold = 0.9497).

Slice 1 - Boundary false positives: different-identity pairs with cosine similarity within ±0.05 of the decision threshold. This slice captures 355 pairs (35.5% of test), of which 91 are actual false acceptances. The high density near the boundary reflects heavily overlapping score distributions: same-person and different-person means are only 0.021 apart (0.936 vs. 0.915). Representative examples (Appendix, Figure 4) show that falsely accepted pairs tend to share pose, lighting, and demographic attributes, producing near-identical pixel patterns despite being different people. A learned feature extractor capturing identity-specific geometry rather than raw luminance would widen this margin.

Slice 2 - High-variation false negatives: same-identity pairs in the bottom 20th percentile of same-person scores (score ≤ 0.9131), totalling 100 pairs (10.0%). Examples (Appendix, Figure 5) show the same individual under dramatically different head pose, lighting, expression, or accessories. Because the verifier operates on raw pixel intensities, even a moderate rotation destroys the element-wise alignment cosine similarity relies on. Pose normalisation or a learned invariant representation would directly address this failure mode.

Section 5: Data-Centric Improvement - Before vs. After

The data-centric intervention was identity capping: limiting each identity to at most 10 pairs and enforcing a 1:1 positive-to-negative ratio. This prevents heavily photographed identities from dominating the pair set and removes the baseline validation class imbalance (1:1.87 → 1:1). Key metrics are compared in Tables 3-4 (Appendix).

The capped system selects a higher threshold (0.9598 vs. 0.9497), yielding a more conservative verifier: FAR drops from 18.2% to 6.4% and precision rises from 69.3% to 78.9%, but FRR climbs from 59% to 76% and balanced accuracy decreases slightly (0.614 → 0.588). AUC is essentially unchanged (0.611 → 0.607), confirming that the binding constraint is the pixel-level representation, not the data composition. The intervention is nonetheless valuable: it exposed that the baseline's 18% FAR was partly an artifact of over-represented identities, and the capped pipeline provides a more honest per-identity evaluation.

Closing

The most important lesson from this iteration loop is that a disciplined evaluation protocol - tracked runs, locked thresholds, and structured error analysis - reveals whether a change genuinely improves a system or merely shifts errors between categories, a distinction that ad hoc experimentation easily obscures.

Appendix

Tables

Table 1

Split	Pairs	Positives	Negatives	Role
Val	457	159 (35%)	298 (65%)	Threshold selection only
Test	1000	500 (50%)	500 (50%)	Final held out reporting

Table 2

Rule	Threshold	Val Balanced Acc	Val F1	Val EER
Maximize balanced accuracy	**0.9497**	**0.5899**	0.4291	**0.415**
Maximize F1	0.8492	0.5162	**0.5203**	**0.415**
Minimize EER	0.9347	0.5850	0.4905	**0.415**

Table 3

	Baseline val	Capped val
Total pairs	457	318
Positive : Negative	159 : 298 (1 : 1.87)	159 : 159 (1 : 1)

Table 4

Metric	Baseline (run_03)	Identity-capped (run_05)	Delta
Threshold	0.9497	0.9598	+0.0101
Balanced accuracy	0.614	0.588	-0.026
F1 score	0.515	0.368	-0.147
AUC (val sweep)	0.611	0.607	-0.004
FAR	18.2%	6.4%	-11.8pp
FRR	59.0%	76.0%	+17.0pp
Precision	69.3%	78.9%	+9.6pp

Figures

Figure 1

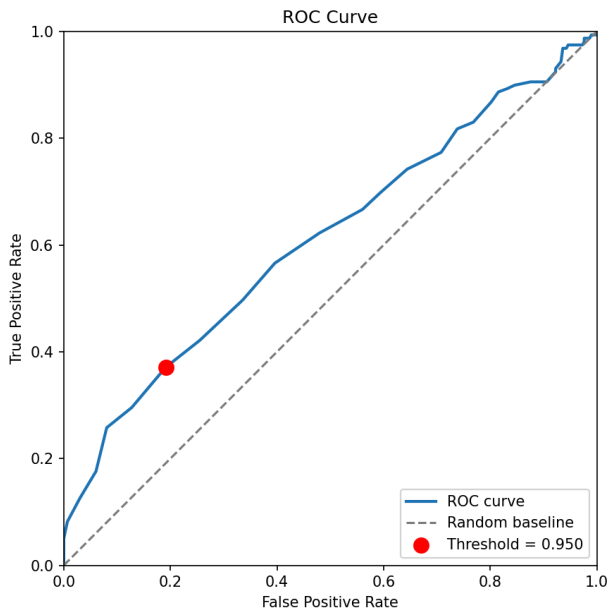
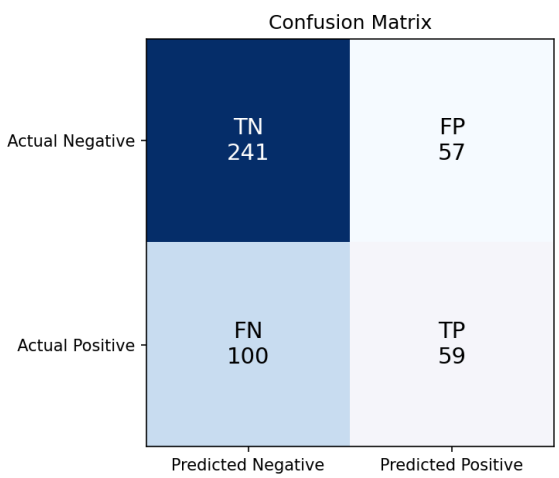


Figure 2

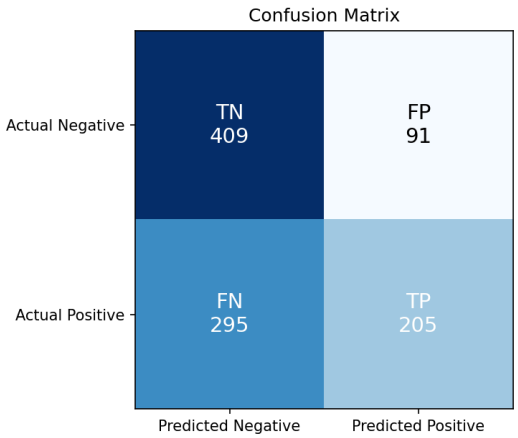
Validation split: run_02



Metric	Value
Balanced accuracy	0.5899
F1 score	0.4291
EER	0.415
FAR (FP/ total negatives)	57/298 = 19.1%
FRR (FN/ total positives)	100/159 = 62.9%

Figure 3

Test split: run_03 (final, held-out)



Metric	Value
Balanced accuracy	0.614
F1 score	0.515
EER	0.372
FAR (FP/ total negatives)	91/500 = 18.2%
FRR (FN/ total positives)	295/500 = 59.0%

Figure 4

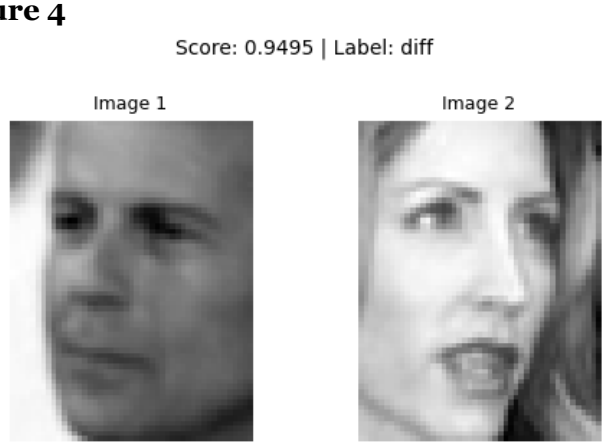


Figure 5